NGC6397

MEMBERSHIP

Stars in catalog	1207
with Gaia GSP-Phot	688 (57%)
with radial velocity	17
Stream id (sID)	69

GEOMETRY · great-circle fit

Centroid RA, Dec	278.2, -53.6 deg
Centroid l, b	341.8, -18.4 deg
Pole RA, Dec	269.9, 35.9 deg
Length (φ1, 2–98%)	26.3 deg
Width σ(φ2), robust	0.97 deg

KINEMATICS

pmRA* med ± σ	4.56 ± 1.93 mas/yr
pmDE med ± σ	-15.96 ± 2.48 mas/yr
μφ1* med ± σ	3.36 ± 1.57 mas/yr
μφ2 med ± σ	-16.37 ± 1.53 mas/yr
Parallax (cat.) med	0.345 ± 0.197 mas
Parallax (wtd, ZP-corr.)	0.382 ± 0.010 mas
Distance	d = 2.6 ± 0.1 kpc
Physical length	1.20 kpc
RV median ± σ(MAD)	21.7 ± 8.8 km/s
RV gradient dV/dφ1	2.44 km/s/deg

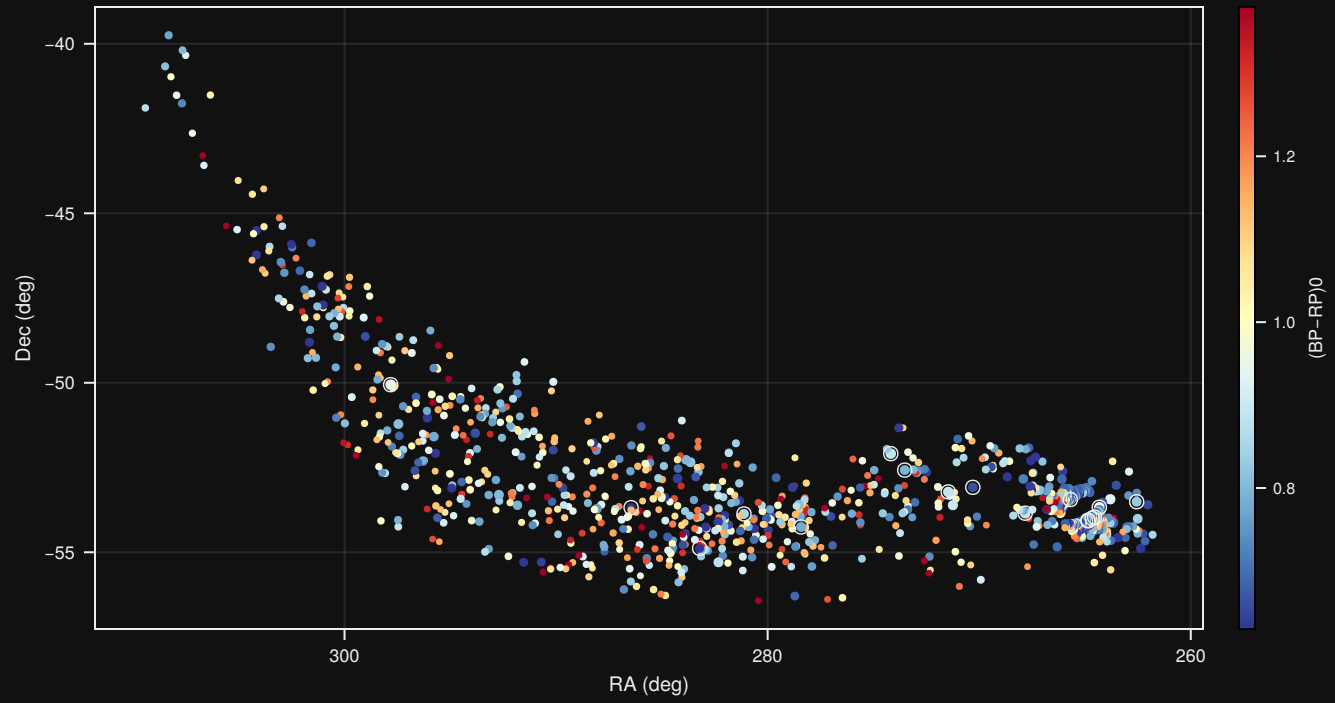
PHOTOMETRY · POPULATIONS

G0 range (median)	12.1–20.0 (18.4)
(BP−RP)0 median	0.95
GSP-Phot [M/H] (n=688)	-2.13 ± 0.93
GSP-Phot Teff median	5367 K
RUWE > 1.4	0.7 %
PM clusters (largest)	80 (37 stars)

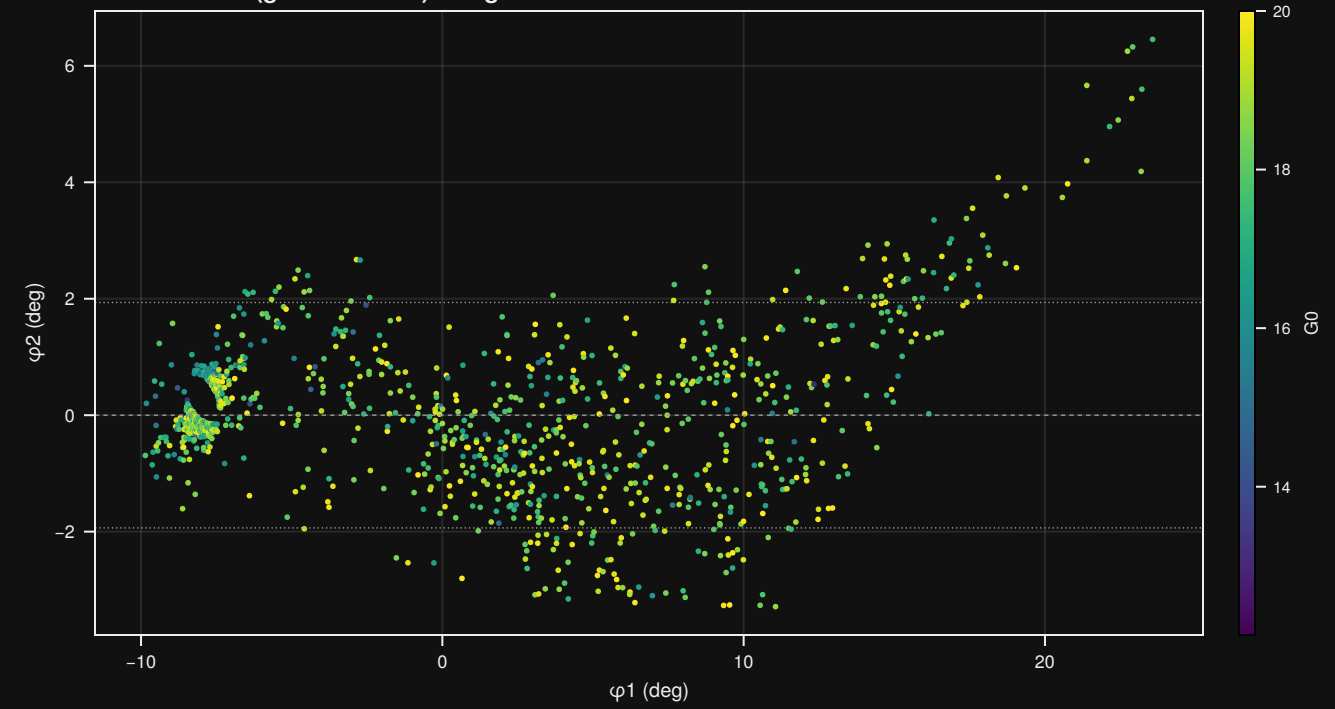
IBATA+2024 · TABLE 3

n catalog | paper	1207 | 1207
n_RV catalog | paper	17 | 17
<[Fe/H]> paper	-2.02
Stellar mass (paper)	2500 Msun
Pole offset vs paper	0.9 deg

Sky map · catalog members (n = 1207) · size ∝ brightness · ring = has RV



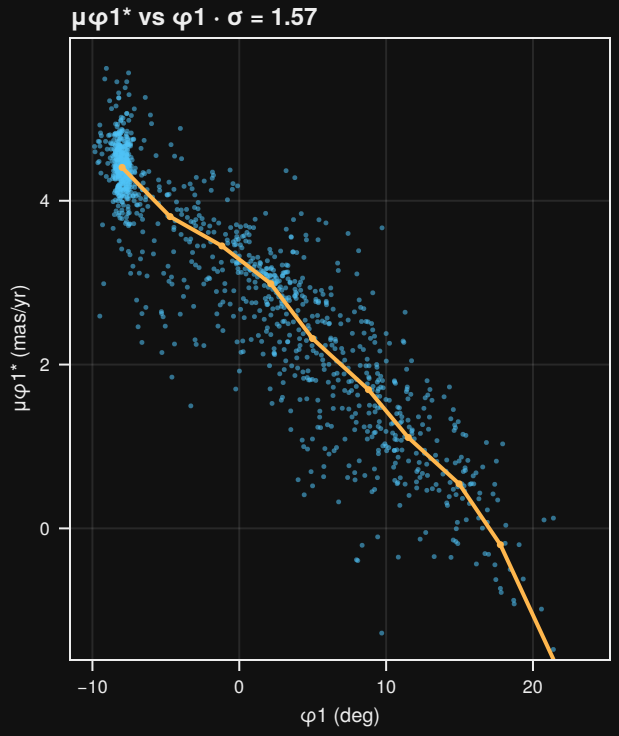
Stream frame (great-circle fit) · length 26.3° · width σ 0.97°

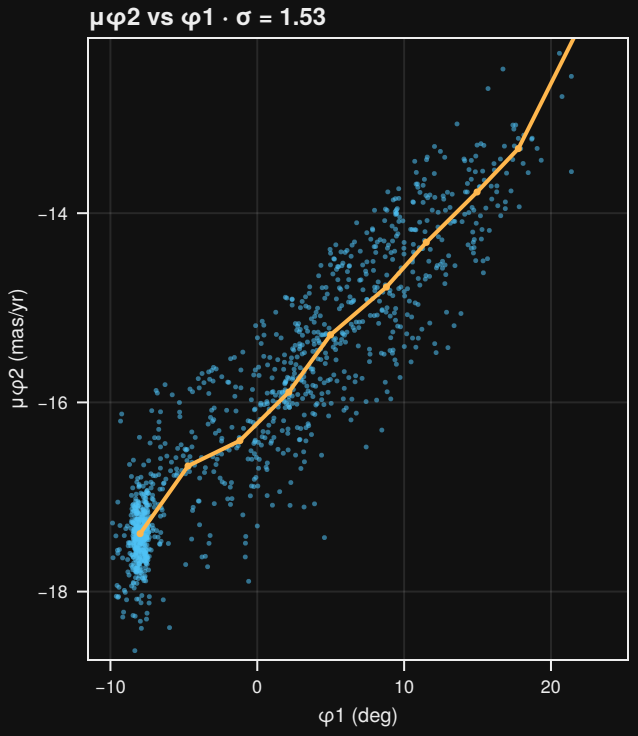


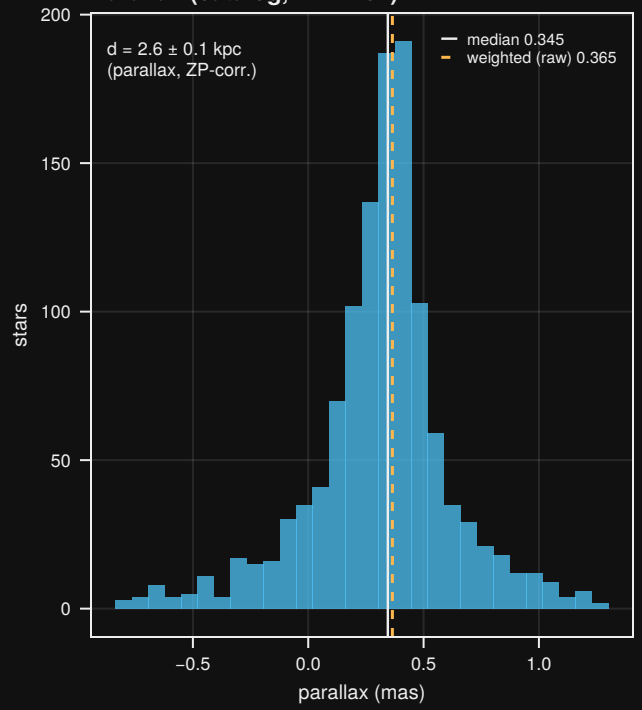
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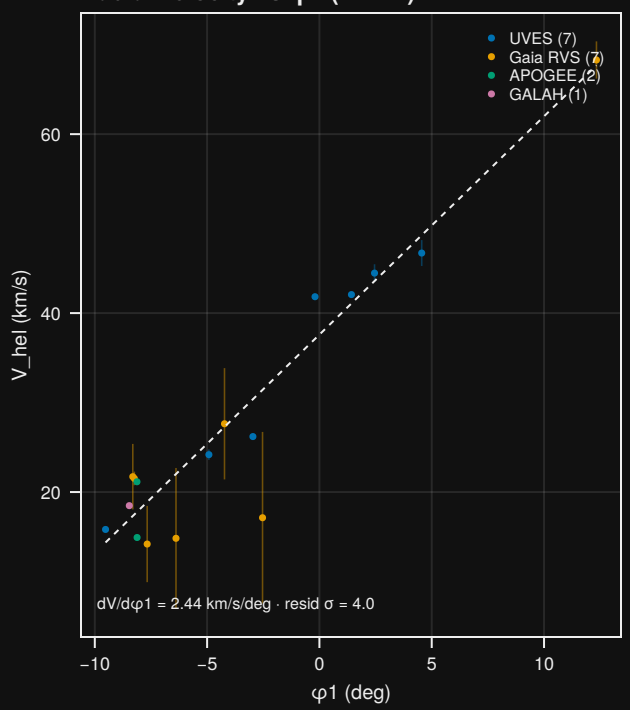
Kinematics · proper motion · parallax · radial velocity

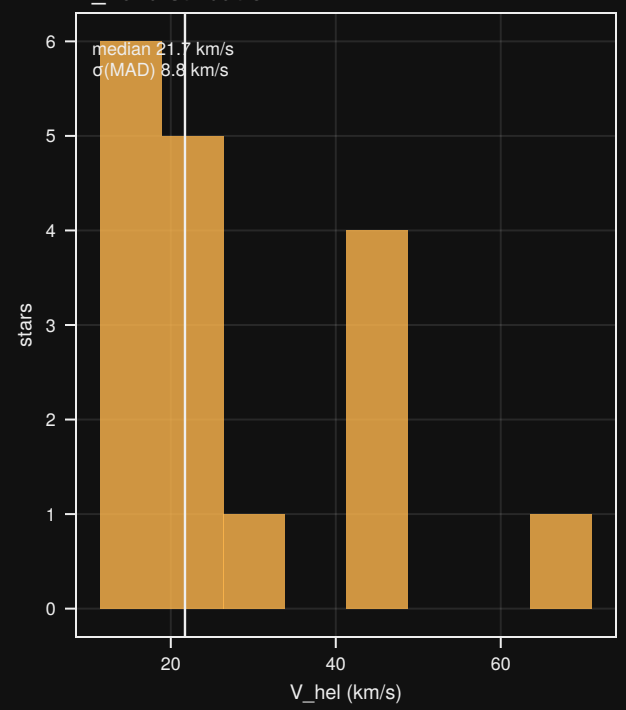






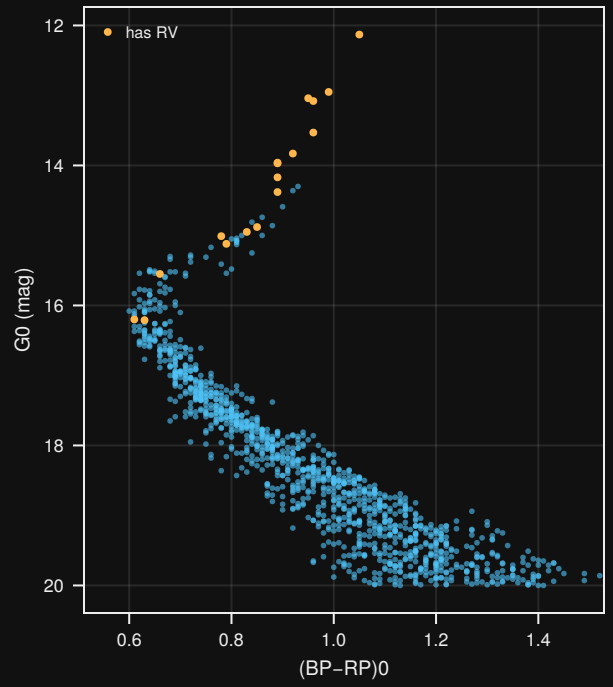




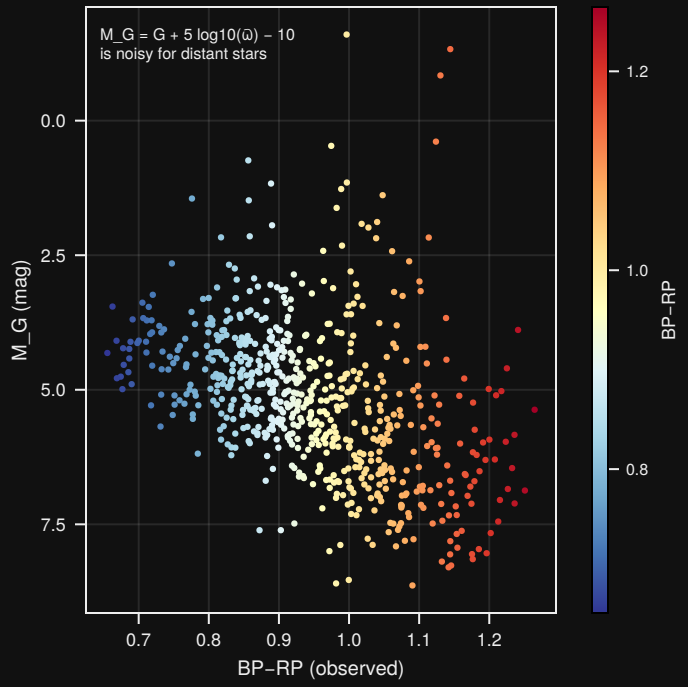


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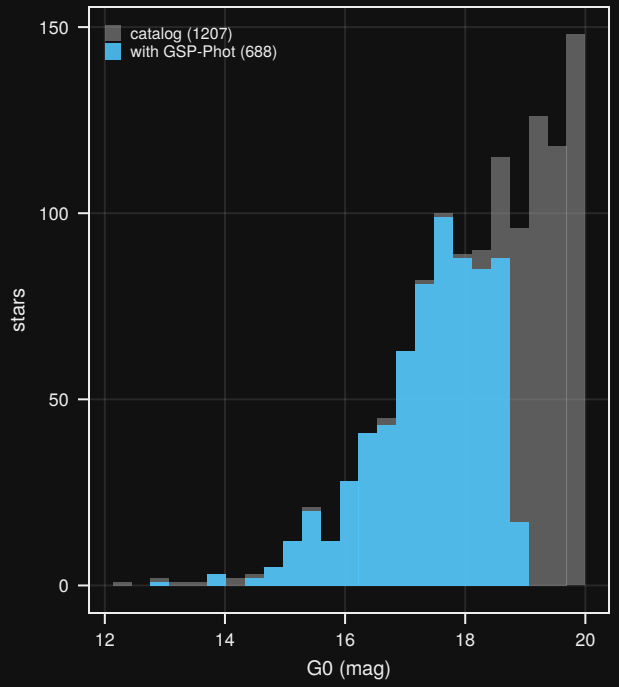
CMD · dereddened (n = 1207)



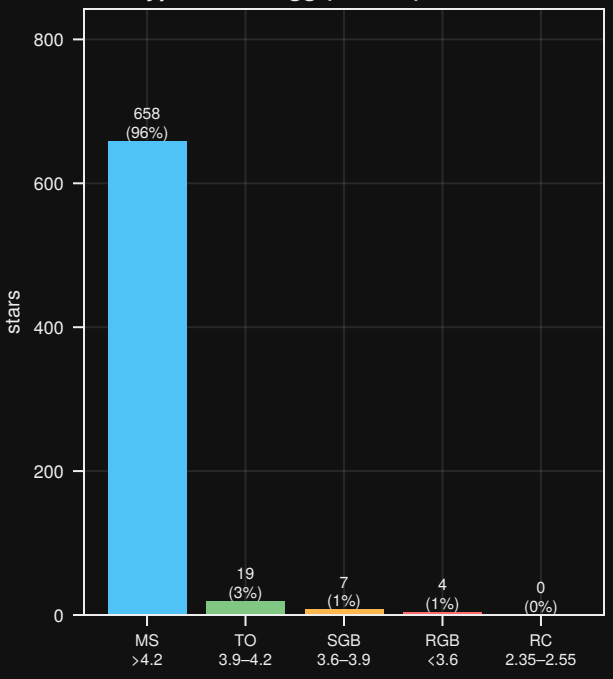
HR diagram · ϖ>0 (n = 673/688)



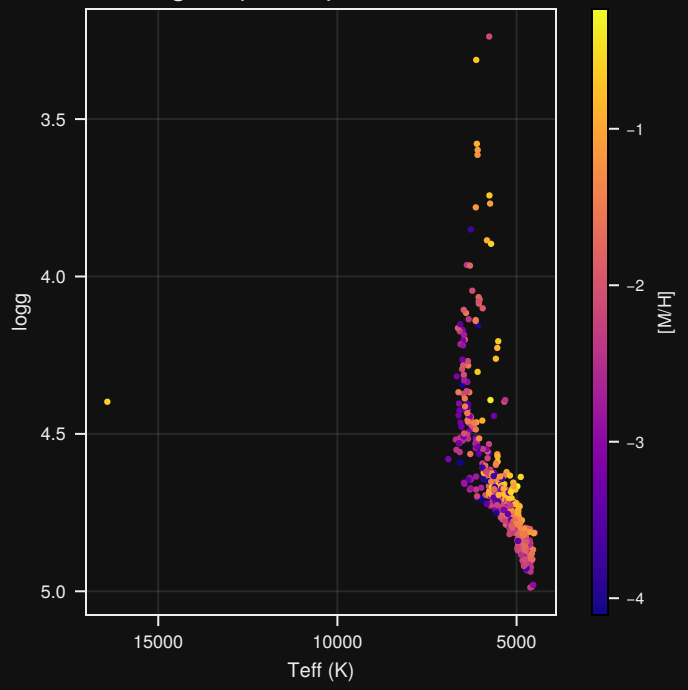
G0 & GSP-Phot coverage



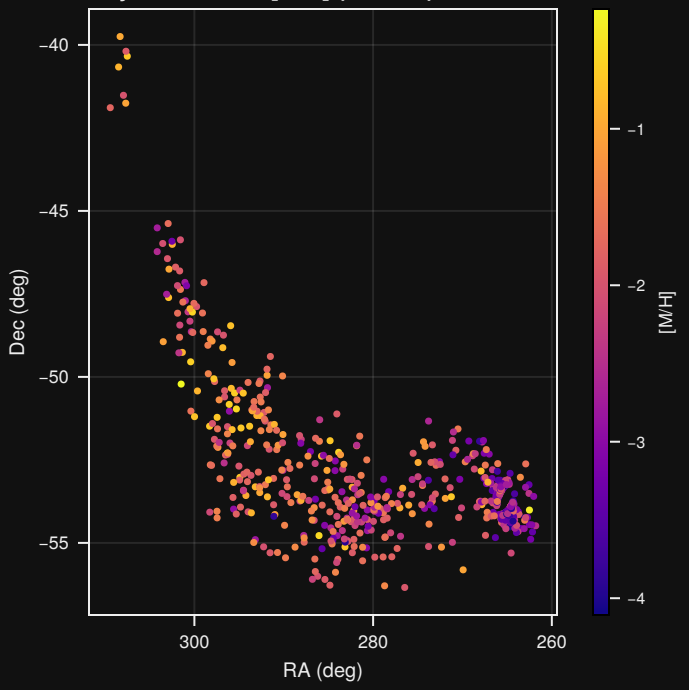
Star types from logg (n = 688)



Kiel diagram (n = 688)



Sky · GSP-Phot [M/H] (n = 688)

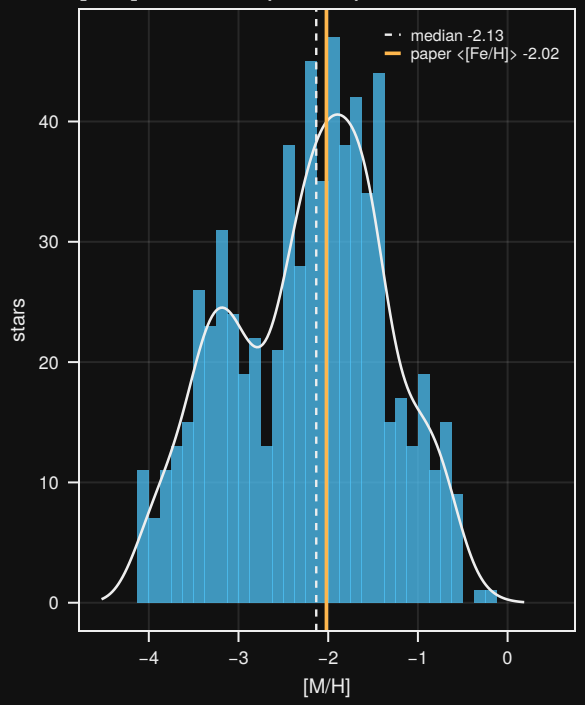


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

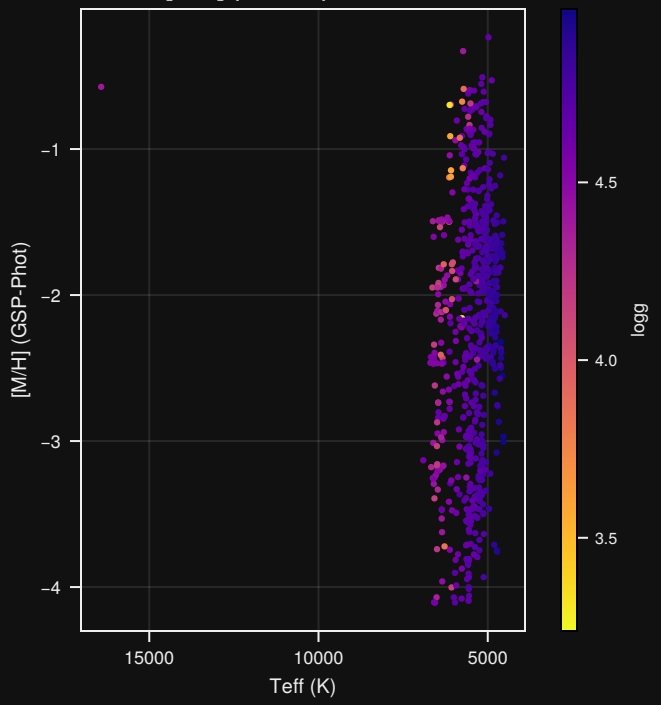
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Metallicity · age · astrometric quality

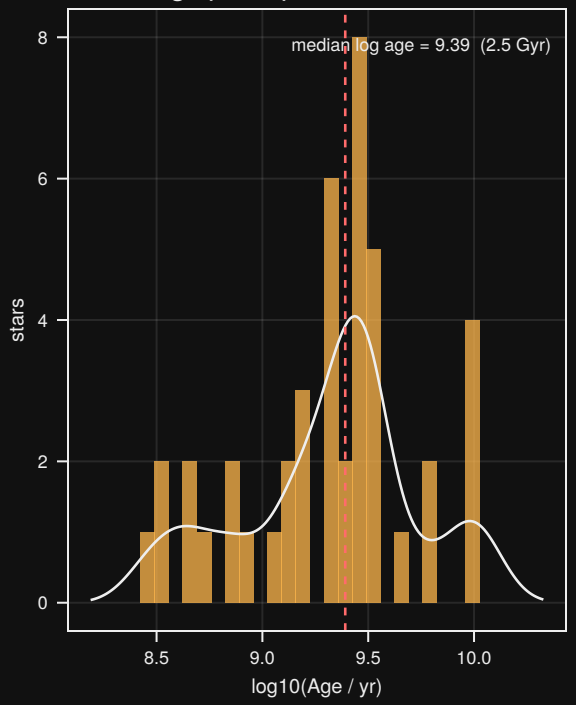
[M/H] GSP-Phot (n = 688)



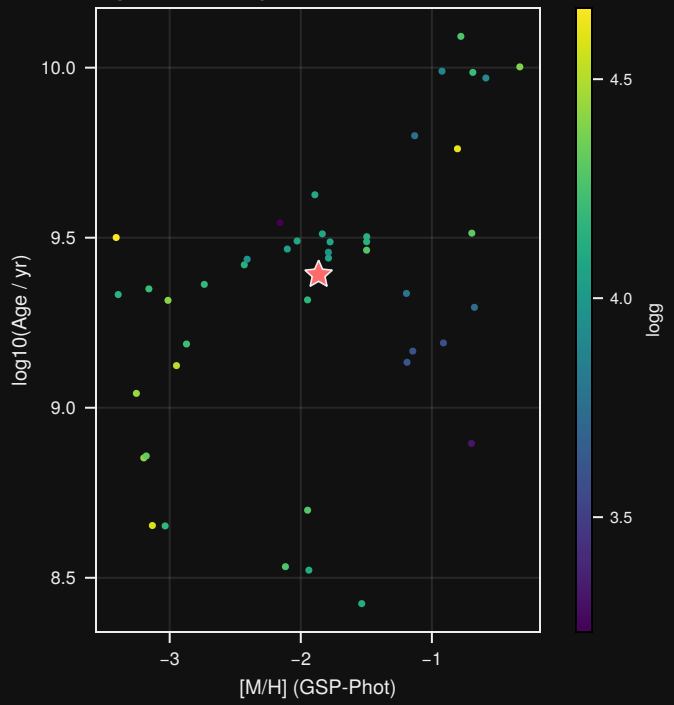
Teff vs [M/H] (n = 688)



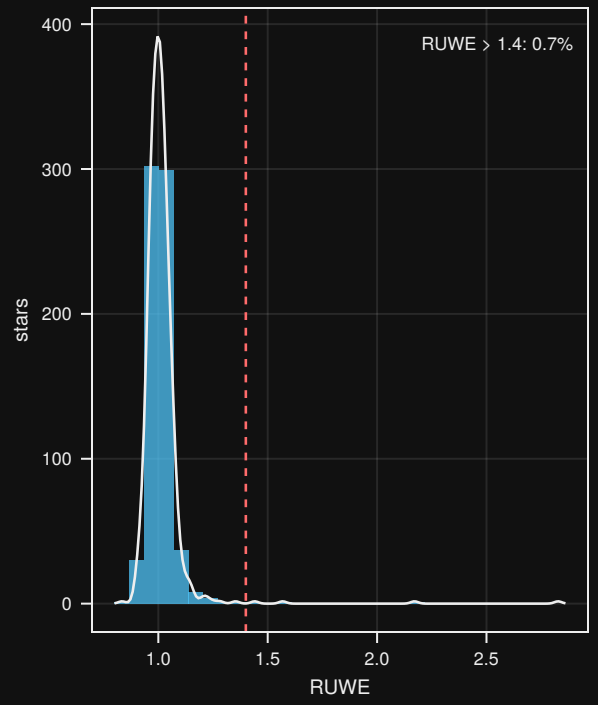
FLAME age (n = 44)



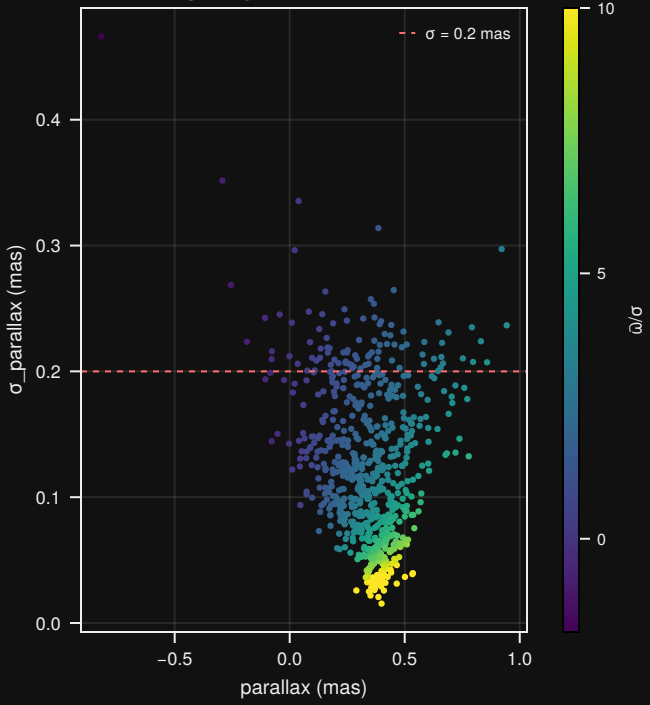
Age–metallicity (n = 44)



RUWE (n = 687)

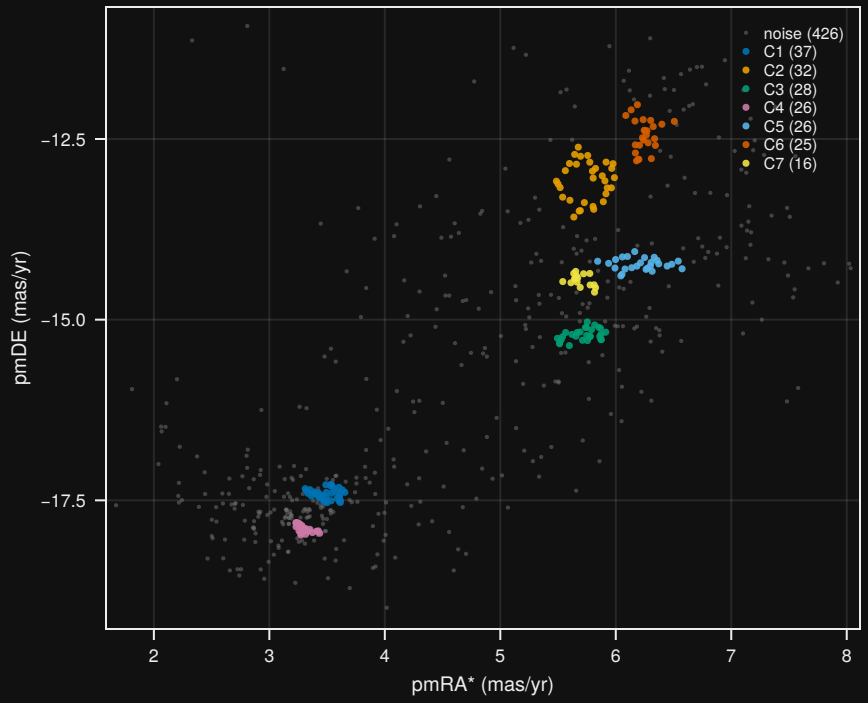


Parallax quality (n = 688)

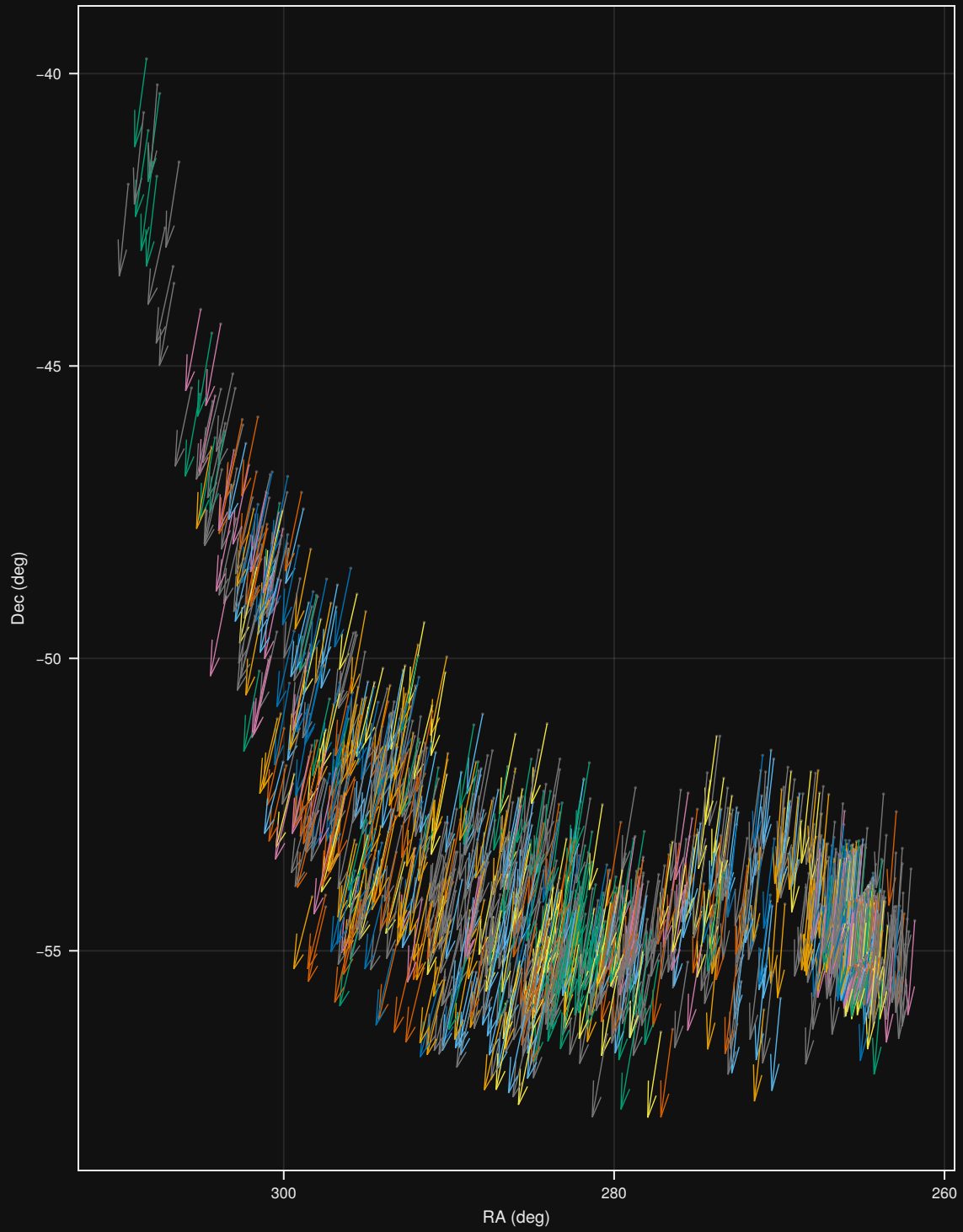


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PM clustering · HDBSCAN · 80 cluster(s) · noise 35%



Proper-motion directions (arrows) coloured by cluster



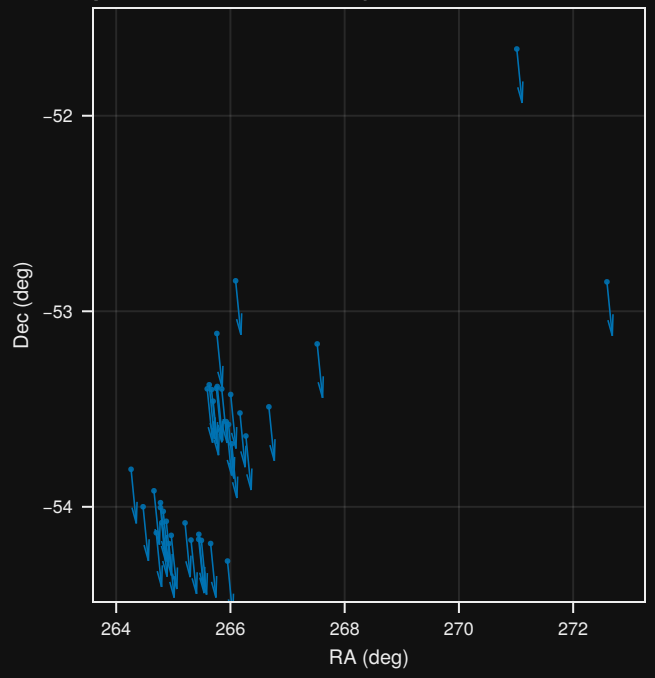
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	426	4.38	-16.18	1.84	-2.10	5
C1	37	3.49	-17.41	0.11	-3.11	0
C2	32	5.77	-13.06	0.19	-1.01	0
C3	28	5.75	-15.22	0.16	-1.96	1
C4	26	3.28	-17.90	0.04	-3.00	0
C5	26	6.23	-14.23	0.20	-1.68	0
C6	25	6.24	-12.44	0.09	-1.81	0
C7	16	5.67	-14.45	0.06	-1.86	0
C8	16	6.28	-13.02	0.09	-1.73	0

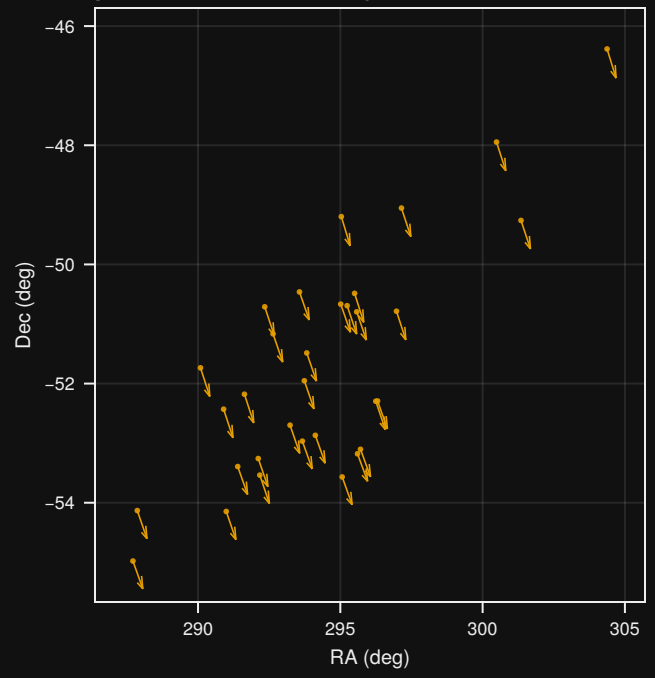
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Per-cluster proper-motion vector maps

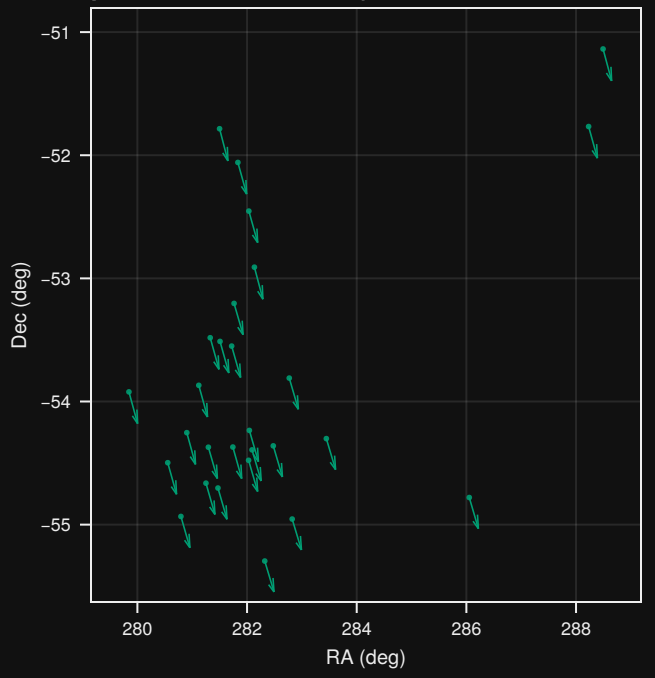
Cluster C1 · n = 37
pm = (3.49, -17.41) mas/yr



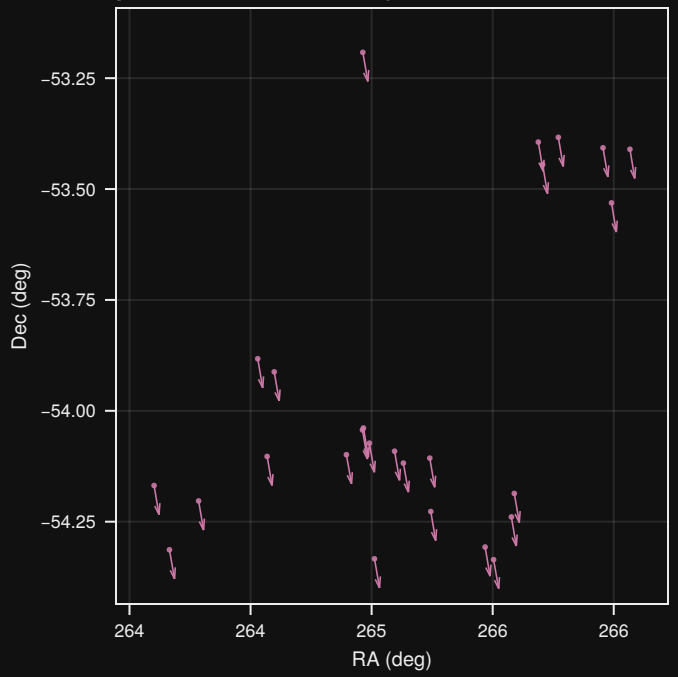
Cluster C2 · n = 32
pm = (5.77, -13.06) mas/yr



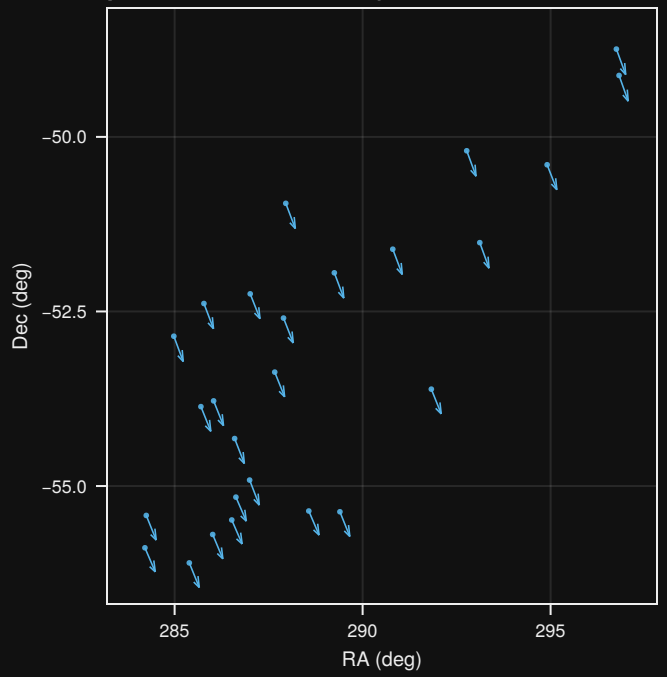
Cluster C3 · n = 28
pm = (5.75, -15.22) mas/yr



Cluster C4 · n = 26
pm = (3.28, -17.90) mas/yr



Cluster C5 · n = 26
pm = (6.23, -14.23) mas/yr



Noise · n = 426

